Chapter 8, Problem 27P

Problem

A branch voltage in an *RLC* circuit is described by

$$\frac{d^2v}{dt^2} + 4\frac{dv}{dt} + 8v = 24$$

If the initial conditions are $v(0) = 0 = dv(0)/dt$, find $v(t)$.

Step-by-step solution

Step 1 of 5

Write the differential equation that describes series *RLC* circuit.

$$\frac{d^2v}{dt^2}+4\frac{dv}{dt}+8v=24$$

Solve this differential equation to find the voltage, $v(t)$.

The complete response of the circuit is,

$$v(t)=v_t(t)+v_{ss}(t)$$

Here,

$v_t(t)$ is the transient or natural response

$v_{ss}(t)$ is the steady-state response.

Find the natural response.

The characteristic equation is,

$$s^2+4s+8=0$$

The roots of the characteristic equation are,

$$\begin{aligned} s_{1,2} &= \frac{-b\pm\sqrt{b^2-4ac}}{2a} \\ &= \frac{-4\pm\sqrt{4^2-(4)(1)(8)}}{2(1)} \\ &= \frac{-4\pm\sqrt{16-32}}{2} \\ &= -2\pm j2 \end{aligned}$$

Step 2 of 5

The roots are complex. Therefore, the transient response is,

$$v(t)=e^{-\alpha t}\left(B_1\cos\omega_d t+B_2\sin\omega_d t\right)$$

The standard form of characteristic roots are,

$$s=-\alpha\pm j\omega_d$$

Therefore,

$$\begin{aligned} \alpha &= 2 \\ \omega_d &= 2 \end{aligned}$$

The natural response is,

$$v(t)=e^{-2t}\left(B_1\cos 2t+B_2\sin 2t\right) \quad \ldots\ldots (1)$$

Step 3 of 5

The steady state response is,

$$v_{ss} = v(\infty)$$

Assume that the steady-state response is constant, K and substitute it in the differential equation.

$$\frac{d^2}{dt^2}(K)+4\frac{d}{dt}(K)+8K = 24$$
$$8K = 24$$
$$K = 3$$

Therefore,

$$v_{ss} = 3 \text{ V}$$

Therefore, the total response is,

$$v(t) = 3+e^{-2t}\left(B_1 \cos 2t + B_2 \sin 2t\right) \quad \text{......} (2)$$

Use the initial conditions to determine the values of B_1 and B_2.

Substitute 0 for t in the equation of $v(t)$.

$$v(0) = 3+e^{-2(0)}\left[B_1 \cos 2(0) + B_2 \sin 2(0)\right]$$
$$v(0) = 3+B_1$$

Substitute 0 for $v(0)$.

$$0 = 3+B_1$$
$$B_1 = -3$$

Step 4 of 5

Recall equation (2).

$$v(t) = 3+e^{-2t}\left(B_1 \cos 2t + B_2 \sin 2t\right)$$

Differentiate equation with respect to *t*.

$$\frac{dv(t)}{dt} = -2e^{-2t}\left(B_1 \cos 2t + B_2 \sin 2t\right)+e^{-2t}\left(-2B_1 \sin 2t + 2B_2 \cos 2t\right)$$

Substitute 0 for t.

$$\frac{dv(0)}{dt} = -2e^{-2(0)}\left[B_1 \cos 2(0) + B_2 \sin 2(0)\right]+e^{-2(0)}\left[-2B_1 \sin 2(0) + 2B_2 \cos 2(0)\right]$$
$$= -2B_1 + 2B_2$$

Substitute 0 for $\frac{dv(0)}{dt}$ and -3 for B_1.

$$0 = -2(-3)+2B_2$$
$$0 = 6+2B_2$$
$$B_2 = -3$$

Step 5 of 5

Now calculate the voltage expression.

Recall equation (2).

$$v(t)=3+e^{-2t}\left(B_1\cos 2t+B_2\sin 2t\right)$$

Substitute -3 for B_1 and -3 for B_2.

$$\begin{aligned}v(t)&=3-3e^{-2t}\cos 2t-3e^{-2t}\sin 2t\\&=3-3\left(\cos 2t+\sin 2t\right)e^{-2t}\\&=3\left[1-\left(\cos 2t+\sin 2t\right)e^{-2t}\right]\text{ V}\end{aligned}$$

Thus, the expression for the voltage $v(t)$ is $\boxed{3\left[1-\left(\cos 2t+\sin 2t\right)e^{-2t}\right]u(t)\text{ V}}$.